Unit 1, Lesson 4: Dilations and Partitions

Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

Learning Targets

- ☐ I can dilate a line segment.
- ☐ I can define midpoint using dilation and understand partitions.
- ☐ I can relate scale factor to dilation.

1.4.1 Warm-Up: Floyd Norman

Floyd Norman was named a Disney Legend in 2007. In 1957, Norman was employed as an inbetweener on Sleeping Beauty. An inbetweener is an animator who draws the intermediate frames between two key frames. The inbetweener's drawings create the illusion of motion.

When asked, "What is the key to growing as an artist?" Norman said, "As a student, I think the most important thing is having a willingness to learn."

1. On a scale of 1 to 10, how willing are you to learn?

When asked about his career Norman stated, "Making mistakes is how we grow, how we learn." He said he had no confidence, but at least he made the first step.

2. Describe a time in your life when you had no confidence, but you made the first step and learned from making mistakes.

Norman described a time when Walt Disney asked him to write. Norman made these two statements as he reflected on doing something he did not think he could do.

- "The only obstacle you face is yourself."
- "If you are willing to do the work, make that commitment, then I don't believe there is an obstacle."

3. What subject(s) are obstacles for you?

1.4.2 Exploration: Dilating a Line Segment

1. The diagram shows points W, W', and W''.

   a. What do you notice about the points?

   b. Estimate the number of times the distance between W and W' will fit between W' and W''.

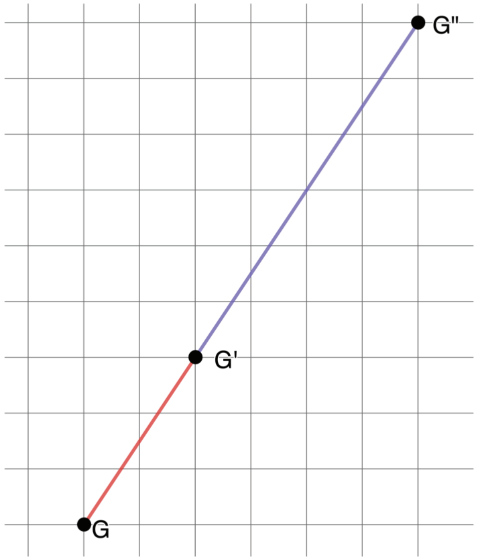
c. Two students described the relationship $\overline{WW''}$ has with $\overline{WW'}$.

Jaquaze	Andrés
$\overline{WW''}$ is three times as long as $\overline{WW'}$.	$\overline{WW''}$ is four times as long as $\overline{WW'}$.

Discuss with your partner which student is correct and the mistake the other student made. Write a brief note to the student who made the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

2. The diagram shows points G, G', and G''.



a. Use patty paper to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$.

b. Discuss with your partner how to use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Summarize your discussion.

c. Ask a classmate how they would use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Record their description and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your description and summary are correct.

Description	My Summary of Their Reason	Initials

A ***dilation*** is a proportional increase or decrease in size in all directions.

d. Use the definition of dilation to describe why $\overline{GG''}$ is a dilation of $\overline{GG'}$.

e. Fill in the blanks to complete the statements.

G' divides $\overline{GG''}$ into a ratio.

The ratio of $\frac{GG'}{GG''}$ is $\frac{\boxed{}}{\boxed{}}$.

Partition means to separate or to divide. A line segment can be partitioned into smaller segments. A ratio can be used to describe the partition.

f. Use the Pythagorean theorem to complete the table.

Segment	Length
GG'	
$G'G''$	
GG''	

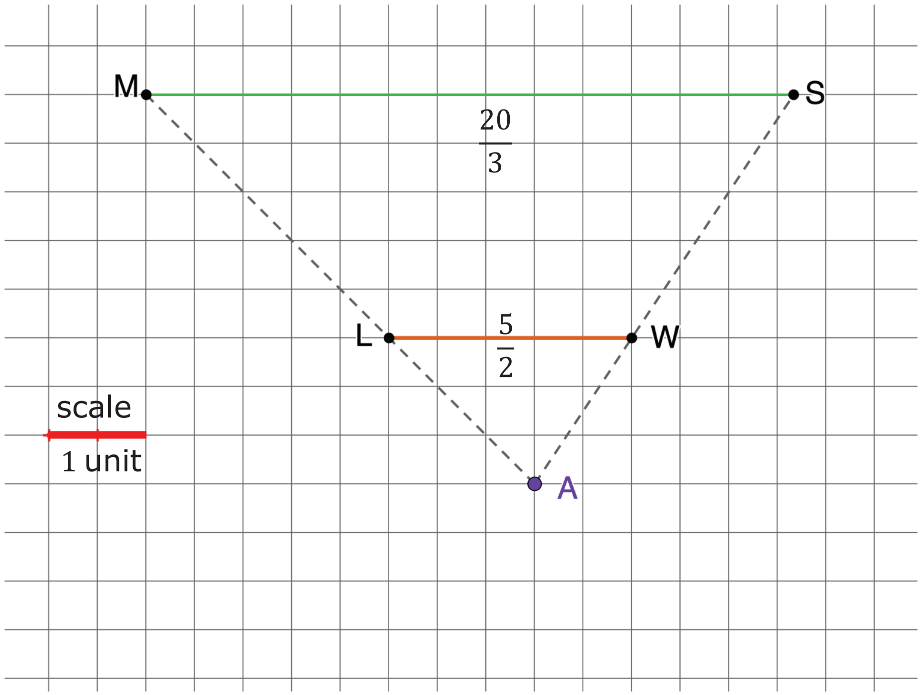
g. Point G' is $\frac{\boxed{}}{\boxed{}}$ of the distance from G to G''.

h. What value can be multiplied to the length of GG' to determine the length of GG''?

The ***scale factor*** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.

1.4.3 Guided Practice: Scale Factor

$\overline{LW}$ is dilated to create $\overline{MS}$. The center of dilation is point A.



1. $\overline{LW}$ has a length of $\frac{5}{2}$ units and $\overline{MS}$ has a length of $\frac{20}{3}$ units. What is the scale factor that was used to create $\overline{MS}$?

2. Use the Pythagorean theorem to complete the table. Round values to the nearest tenth of a unit.

Segment	**Length**	**Segment**	**Length**
$\overline{AL}$		$\overline{AW}$	
$\overline{LM}$		$\overline{WS}$	3.0
$\overline{AM}$		$\overline{AS}$	

3. Point L is $\frac{\square}{\square}$ of the distance from A to M.

4. What value can be multiplied to the length of $\overline{AL}$ to determine the length of $\overline{AM}$?

5. Point W is $\frac{\square}{\square}$ of the distance from A to S.

6. What value can be multiplied to the length of $\overline{AW}$ to determine the length of $\overline{AS}$?

7. What do you notice about each of the following ratios?

   - $LW : MS$
   - $AL : AM$
   - $AW : AS$

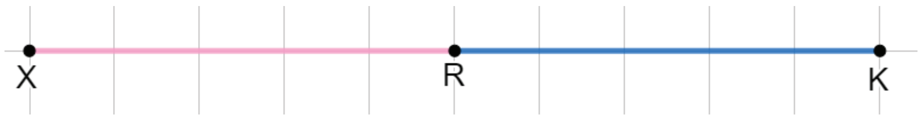
8. Work with your partner to explain how to find the scale factor of a dilation.

1.4.4 Bring It Together: Determining Scale Factor

As the class shares strategies for finding scale factor, choose a strategy other than yours to describe.

Strategy	I Liked the Strategy Because…

1. $\overline{XR}$ and $\overline{RK}$ are shown.



   a. What is the scale factor that can be used to dilate $\overline{XR}$ to create $\overline{XK}$?

   b. What is the scale factor that can be used to dilate $\overline{XK}$ to create $\overline{XR}$?

   c. Describe point R in relation to $\overline{XK}$.

The ***midpoint*** of a line segment is the point that divides a line segment into two equal segments.

1.4.5 Wrap-Up: Cool Down

1. Isometry is a transformation that preserves distance and angle measure. Write a short note that describes whether a dilation preserves distance or not.

Unit 1, Lesson 5: Transformations on the Plane – Part 1

Benchmark

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

Learning Targets

- ☐ I can identify the angle of rotation of a transformation of a figure on a plane.
- ☐ I can identify the center of rotation of a transformation of a figure on a plane.
- ☐ I can identify the direction of a rotation of a transformation of a figure on a plane.
- ☐ I can distinguish a rotation from other transformations.